QueryMaster: An End-to-End Text-to-SQL Autonomous Agent

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***Abstract*—This work introduces QueryMaster, an autonomous text-to-SQL agent that automates database queries by utilizing a quantized Large Language Model (LLM). In particular, Low-Rank Adaptation (LoRA) designed for Apple Silicon via the MLX framework is used to refine the Qwen3-8B-4bit model. The model is trained using a reliable combination of the BIRD and SynSQL datasets to guarantee mastery over cross-domain schemas and sophisticated SQL operations. The suggested solution achieves a competitive execution accuracy of 67.4% after being thoroughly tested on the Spider 1.0 development set. Additionally, the model's practical feasibility is successfully demonstrated through the development of a lightweight web interface for real-time local inference. The findings show that highly proficient text-to-SQL agents may be produced by parameter-efficient fine-tuning on consumer-grade hardware, setting the stage for future scalable microservices integration.**

***Index Terms*— Text-to-SQL, Large Language Models, Low-Rank Adaptation, Database Automation.**

I. Introduction

THe relational databases are widely used, they typically require knowledge of Structured Query Language (SQL). This can make it difficult for non-technical users to interact with them. Due to this, the Text-to-SQL task - translating natural language question into executable SQL statements - has emerged as an important research focus over the last few years. Large language models (LLM) have made significant strides toward improving performance on this task; however, most previous solutions have required either high-end computer equipment or rely on cloud services.

QueryMaster is presented here as a local Text-to-SQL Agent designed to operate on a consumers' grade hardware. QueryMaster is built upon the Qwen3-8B Model. It was trained with low-rank adaptation (loRA) in order to allow efficient training under limited hardware. Training data for QueryMaster was generated through the combination of BIRD and SynSQL dataset examples to provide the ability to generalize better to different database schemas and SQL patterns.

Optimization for running on Apple Silicon was completed using the MLX framework. The LoRa adapters created during optimization were then integrated into a lightweight streamlit application to enable local inference. Users input a database schema and natural language query and receive the generated SQL statement in real-time from the QueryMaster system.

Evaluation was done against the Spider 1.0 Development Set. Accuracy of execution for the model was at 67.4%. Results indicate that a working Text-to-SQL System can be implemented and deployed locally utilizing parameter-efficient fine tuning without having to use large scale server infrastructure.

II. Methodology

A. Data Engineering

A unique dataset was created by combining the BIRD and SynSQL datasets in order to train a reliable model that could handle a variety of database systems. This combination exposes the model to cross-domain schemas, nested subqueries, and intricate JOIN procedures. A ChatML format appropriate for conversational instruction tweaking was created from the raw data.

B. Model and Hardware Optimization

The base model selected for this project is Qwen3-8B, quantized to 4-bit precision. To overcome the memory constraints of local training, Low-Rank Adaptation (LoRA) was implemented. The training pipeline was specifically optimized for Apple Silicon (Unified Memory Architecture) using the MLX framework, preventing Out-Of-Memory (OOM) errors and significantly accelerating the training speed.

C. Prototype Interface

For the midterm demonstration, a lightweight web application was developed using the Streamlit framework. This interface allows users to input their database schema (DDL) and natural language queries, enabling real-time, local inference testing of the fine-tuned adapters.

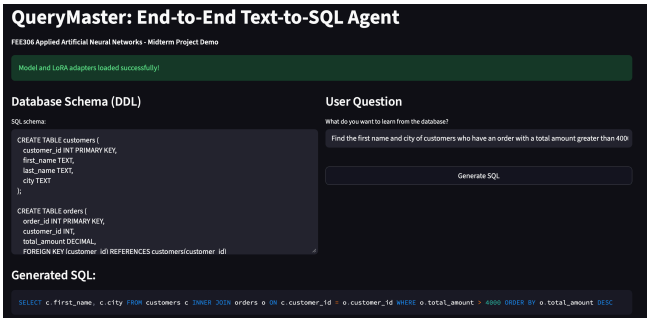

Figure 1: Streamlit web interface for real-time inference.

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III. Results

A. Execution Accuracy

The fine-tuned model was evaluated on the highly competitive Spider 1.0 development set, which contains complex schema layouts and difficult query structures. The model achieved an overall Execution Accuracy of 67.4% (697 out of 1034 queries matched successfully). Additionally, the model reduced syntax errors to just 4.1% while handling schema-heavy prompts with great resilience.

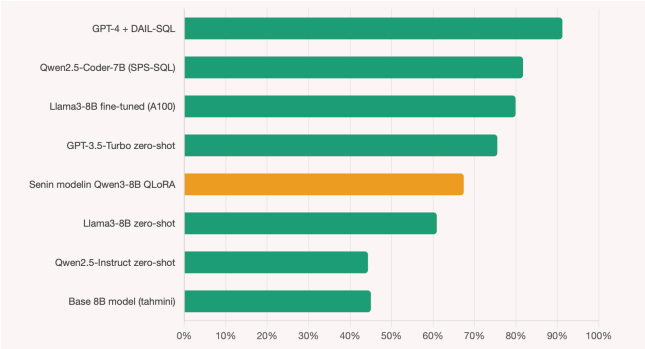

Figure 2: Execution accuracy comparison against baseline LLMs.

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(base) mehmetzahitavci@Mehmet-MacBook-Pro QueryMaster %
  ============================================================
  EVALUATION RESULTS (Spider 1.0 Dev Set)
  ============================================================
    Execution Accuracy : 67.4% (697/1034)
    SQL Syntax Errors  : 42 (4.1%)

    Accuracy by Difficulty:
      Easy        :  80.4% (344/428)
      Medium      :  63.2% (277/438)
      Hard        :  44.0% (51/116)
      Extra Hard  :  48.1% (25/52)
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Figure 3: Execution accuracy breakdown by query difficulty.

B. Inference Performance

Real-time applicability is an important statistic for autonomous agents that goes beyond accuracy. The model demonstrated an amazing generation speed of roughly 40–45 tokens per second during local inference testing using the MLX framework. The system's effectiveness for real-world production contexts was demonstrated by the generation of complicated SQL queries involving INNER JOINs in less than 1.5 seconds.

IV. Conclusion and Future Work

This midterm project effectively showed the feasibility of implementing an autonomous text-to-SQL agent on consumer-grade hardware. The Qwen3-8B model obtained a highly competitive Execution Accuracy of 67.4% on the Spider assessment suite by utilizing the MLX framework and LoRA fine-tuning. The findings show that a robust combination of training datasets (BIRD and SynSQL) and parameter-efficient fine-tuning enable the model to generalize well to complicated, unknown database designs. The created Streamlit prototype effectively verifies the model's capacity for real-time inference.

The ultimate objective of QueryMaster is to function as an enterprise-grade application, even though the current version offers a strong proof-of-concept. Future research will therefore concentrate on converting this design into a microservices pipeline that is completely scalable. A FastAPI service will contain the AI inference engine. To manage safe database executions, a main backend powered by Java and Spring Boot will then be incorporated. Lastly, the existing prototype will be replaced by a dynamic React frontend that will enable non-technical users to see the generated SQL outputs using automated graphs and charts.

References

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